

Practical # 03

Objective:

Apply multi-dimensional arrays to construct and test matrix operations.

Theory:

A multi-dimensional array is an array of arrays, commonly used to represent matrices and tables. The most common is the two-dimensional array (matrix), which stores data in rows and columns.

In this lab, students will learn how to:

- Declare and initialize two-dimensional arrays.
- Traverse a matrix using nested loops.
- Perform basic matrix operations such as addition, subtraction, transpose, and multiplication.
- Test and display results of these operations.

General form of declaring N-dimensional arrays:

data_type array_name[size1][size2]....[sizeN];

Total number of elements that can be stored in a multidimensional array can be calculated by multiplying the size of all the dimensions.

Example program: *Declare and print elements of multi-dimensional Array.*

```
#include<iostream>
using namespace std;
int main()
{
    // an array with 3 rows and 2 columns.
    int x[3][2] = {{0,1}, {2,3}, {4,5}};
    // output each array element's value
    for (int i = 0; i < 3; i++)
    {
        for (int j = 0; j < 2; j++)
        {
            cout << "Element at x[" << i
                << "][" << j << "]: ";
            cout << x[i][j]<<endl;
        }
    }
    return 0; }
```

OUTPUT

```
Element at x[0][0]: 0
Element at x[0][1]: 1
Element at x[1][0]: 2
Element at x[1][1]: 3
Element at x[2][0]: 4
Element at x[2][1]: 5
```

Review Questions/ Exercise:

1. **Construct** a 2D array to store elements of a matrix and **implement** traversal to print all elements.
2. **Implement** a program to calculate the sum of diagonal elements of a square matrix.
3. **Develop** a program to find the largest and smallest elements in a matrix.
4. **Construct** a program to check whether a given matrix is symmetric.
5. **Develop** a program to calculate the sum of each row and each column of a matrix.
6. **Implement** a program to rotate a matrix by 90 degrees.

Name: _____

Roll #: _____

Date: _____

Subject Teacher